Module 1: Crystal Physics & Reciprocal Lattice

Complete Worked Solutions & TA Rubric

PH7302E Condensed Matter Physics
Department of Physics
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Preface for Teaching Assistants

This document serves as a comprehensive solution rubric and grading guide for Module 1 of PH7302E Condensed Matter Physics. It contains:

1. Complete solutions to all assignment problems
2. Detailed grading rubrics with point allocations
3. Common student errors and misconceptions
4. TA guidelines for consistent grading
5. Supplementary teaching materials

Total Points for Assignment: 100

1 Grading Policy

- Each problem is worth the points indicated
- Partial credit should be awarded for:
  - Correct approach even if final answer is wrong
  - Correct setup of equations
  - Appropriate use of formulas
  - Clear explanation of reasoning
- Deductions should be made for:
  - Missing units
  - Incorrect significant figures
  - Lack of clarity in derivations
  - Missing steps in proofs

2 Assignment 1: Complete Solutions

Lecture 1 Problems

Problem 1: Packing Fraction of FCC (5 points)

Solution: In FCC, atoms touch along face diagonals. If a is the lattice constant and r is the atomic radius:

$$4r = a\sqrt{2} \implies r = \frac{a\sqrt{2}}{4}$$

Volume of atoms per unit cell:

- 4 atoms per unit cell
- Volume of one atom $= \frac{4}{3}\pi r^3$
- Total atomic volume $= 4 \times \frac{4}{3}\pi r^3 = \frac{16}{3}\pi r^3$

Packing fraction:

$$\eta = \frac{4 \times \frac{4}{3}\pi r^3}{a^3} = \frac{16\pi r^3}{3a^3} = \frac{16\pi}{3}\left(\frac{a\sqrt{2}}{4}\right)^3 \frac{1}{a^3}$$

$$\eta = \frac{16\pi}{3} \times \frac{2\sqrt{2}}{64} = \frac{32\sqrt{2}\pi}{192} = \frac{\pi}{3\sqrt{2}} \approx 0.7405$$

Grading Rubric:

Criterion	Points
Correct relationship between r and a ($4r = a\sqrt{2}$)	2
Correct number of atoms per unit cell (4)	1
Correct formula for packing fraction	1
Correct final result ($\pi/3\sqrt{2} \approx 0.74$)	1

Common Errors:

- Using incorrect relationship between r and a
- Forgetting to multiply by number of atoms
- Using wrong number of atoms per unit cell

Problem 2: Atoms in Diamond Conventional Cell (3 points)

Solution: Diamond structure: FCC lattice with 2-atom basis at (0,0,0) and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$.

FCC has 4 lattice points per conventional cell. Each lattice point has 2 atoms associated with it. Total atoms $= 4 \times 2 = 8$ atoms per conventional cell.

Grading Rubric:

Criterion	Points
Recognizes FCC with 4 lattice points per cell	1
Recognizes 2-atom basis	1
Correct final answer (8 atoms)	1

Problem 3: (110) Plane in BCC Lattice (4 points)

Solution: The (110) plane in a BCC lattice contains:

- Corner atoms at (0,0,0), (1,0,0), (0,1,0), (1,1,0) in the plane
- Body center atoms at $(\frac{1}{2}, \frac{1}{2}, 0)$ in the plane
- The plane cuts through the body-centered atom

Grading Rubric:

Criterion	Points
Correct identification of plane intercepts	2
Correct identification of atoms in plane	1
Clear sketch with labeling	1

Problem 4: Miller Indices from Intercepts (4 points)

Solution: Intercepts: $2a, 3b, 6c$

1. Intercepts in units of lattice constants: $(2, 3, 6)$
2. Take reciprocals: $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$
3. Clear fractions (multiply by 6): $(3, 2, 1)$
4. Miller indices: (321)

Grading Rubric:

Criterion	Points
Correct intercepts in units of lattice constants	1
Correct reciprocals	1
Correct clearing of fractions	1
Correct final Miller indices	1

Problem 5: Interplanar Spacing for (220) Planes in Copper (4 points)

Solution: For cubic lattice:

$$d_{220} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{3.61}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{3.61}{\sqrt{8}} = \frac{3.61}{2\sqrt{2}} = 1.28 \text{ Å}$$

Grading Rubric:

Criterion	Points
Correct formula for cubic interplanar spacing	1
Correct values for h, k, l	1
Correct arithmetic	1
Correct units	1

Problem 6: $(11\bar{2})$ **Plane (4 points)**

Solution: For the $(11\bar{2})$ plane:

- Intercepts: $(\frac{1}{1}, \frac{1}{1}, \frac{1}{-2}) = (1, 1, -\frac{1}{2})$
- The plane intersects the x-axis at a, y-axis at b, and z-axis at $-c/2$
- The negative intercept means the plane cuts the negative z-axis

Grading Rubric:

Criterion	Points
Correct intercepts from Miller indices	2
Correct drawing with axes labeled	1
Correct indication of negative intercept	1

Lecture 2 Problems

Problem 7: Reciprocal Lattice of 2D Hexagonal Lattice (8 points)

Solution: Given: $\mathbf{a}_1 = a(1, 0)$, $\mathbf{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$

For 2D, using the vector cross product with $\hat{z}$:

$$V_{2D} = |\mathbf{a}_1 \times \mathbf{a}_2| = a^2 \frac{\sqrt{3}}{2}$$

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \hat{z}}{V_{2D}} = 2\pi \frac{a(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0) \times (0, 0, 1)}{a^2\sqrt{3}/2}$$

$$= 2\pi \frac{a(\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0)}{a^2\sqrt{3}/2} = \frac{2\pi}{a}(1, -\frac{1}{\sqrt{3}})$$

$$\mathbf{b}_2 = 2\pi \frac{\hat{z} \times \mathbf{a}_1}{V_{2D}} = 2\pi \frac{(0, 0, 1) \times a(1, 0, 0)}{a^2\sqrt{3}/2} = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}})$$

The angle between $\mathbf{b}_1$ and $\mathbf{b}_2$ is 60°:

$$\cos\theta = \frac{\mathbf{b}_1 \cdot \mathbf{b}_2}{|\mathbf{b}_1||\mathbf{b}_2|} = \frac{2\pi/a \times 0 + (-2\pi/(a\sqrt{3})) \times (4\pi/(a\sqrt{3}))}{(2\pi/a)(2\pi/a)(2/\sqrt{3})} = \frac{1}{2}$$

Therefore $\theta = 60°$, confirming hexagonal symmetry.

Grading Rubric:

Criterion	Points
Correct 2D reciprocal lattice definition	2
Correct calculation of V_{2D}	1
Correct calculation of $\mathbf{b}_1$	1.5
Correct calculation of $\mathbf{b}_2$	1.5
Correct angle calculation	1
Conclusion: reciprocal lattice is hexagonal	1

Problem 8: Reciprocal of BCC is FCC (8 points)

Solution: Given BCC direct primitive vectors:

$$\mathbf{a}_1 = \frac{a}{2}(-\hat{x} + \hat{y} + \hat{z}), \quad \mathbf{a}_2 = \frac{a}{2}(\hat{x} - \hat{y} + \hat{z}), \quad \mathbf{a}_3 = \frac{a}{2}(\hat{x} + \hat{y} - \hat{z})$$

$$\begin{aligned}
\mathbf{a}_2 \times \mathbf{a}_3 &= \frac{a^2}{4}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \\
&= \frac{a^2}{4}[\hat{x}((-1)(-1) - (1)(1)) - \hat{y}((1)(-1) - (1)(1)) + \hat{z}((1)(1) - (-1)(1))] \\
&= \frac{a^2}{4}[\hat{x}(1-1) - \hat{y}(-1-1) + \hat{z}(1+1)] \\
&= \frac{a^2}{4}[0\hat{x} + 2\hat{y} + 2\hat{z}] = \frac{a^2}{2}(\hat{y} + \hat{z})
\end{aligned}$$

$V = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) = \frac{a}{2}(-\hat{x} + \hat{y} + \hat{z}) \cdot \frac{a^2}{2}(\hat{y} + \hat{z}) = \frac{a^3}{4}(1+1) = \frac{a^3}{2}$

$$\mathbf{b}_1 = 2\pi\frac{\mathbf{a}_2 \times \mathbf{a}_3}{V} = 2\pi\frac{a^2(\hat{y} + \hat{z})/2}{a^3/2} = \frac{2\pi}{a}(\hat{y} + \hat{z})$$

Similarly:

$$\mathbf{b}_2 = \frac{2\pi}{a}(\hat{x} + \hat{z}), \quad \mathbf{b}_3 = \frac{2\pi}{a}(\hat{x} + \hat{y})$$

These are FCC primitive vectors with lattice constant $4\pi/a$, proving the reciprocal of BCC is FCC.

Grading Rubric:

Criterion	Points
Correct BCC primitive vectors	1
Correct calculation of cross products	2
Correct calculation of volume V	1
Correct calculation of $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$	2
Recognition of FCC structure	1
Correct lattice constant $(4\pi/a)$	1

Problem 9: $\mathbf{G}_{hkl}$ Perpendicular to (hkl) Plane (6 points)

Solution: Given: $\mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$

The (hkl) plane contains vectors such as $\mathbf{a}_1/h - \mathbf{a}_2/k$ (both endpoints lie in the plane).

$$\begin{aligned}\mathbf{G}_{hkl} \cdot (\mathbf{a}_1/h - \mathbf{a}_2/k) &= (h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3) \cdot (\mathbf{a}_1/h - \mathbf{a}_2/k) \\ &= h\mathbf{b}_1 \cdot \mathbf{a}_1/h - h\mathbf{b}_1 \cdot \mathbf{a}_2/k + k\mathbf{b}_2 \cdot \mathbf{a}_1/h - k\mathbf{b}_2 \cdot \mathbf{a}_2/k \\ &\quad + l\mathbf{b}_3 \cdot \mathbf{a}_1/h - l\mathbf{b}_3 \cdot \mathbf{a}_2/k\end{aligned}$$

Using $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$:

$$= 2\pi - 0 + 0 - 2\pi + 0 - 0 = 0$$

Therefore $\mathbf{G}_{hkl}$ is perpendicular to the (hkl) plane.

Grading Rubric:

Criterion	Points
Correct expression for $\mathbf{G}_{hkl}$	1
Correct identification of vectors in the plane	1.5
Correct application of orthogonality relation	1.5
Correct final conclusion ($\mathbf{G} \perp$ plane)	2

Problem 10: Bragg Angle for NaCl (200) (5 points)

Solution: For cubic: $d_{200} = \frac{a}{\sqrt{2^2+0^2+0^2}} = \frac{5.64}{2} = 2.82$ Å

Using Bragg's law: $2d\sin\theta = n\lambda$ For first order ($n = 1$):

$$\sin\theta = \frac{\lambda}{2d} = \frac{1.54}{2 \times 2.82} = \frac{1.54}{5.64} = 0.2730$$

$$\theta = \sin^{-1}(0.2730) = 15.84^\circ$$

$$2\theta = 31.68^\circ$$

Grading Rubric:

Criterion	Points
Correct d_{200} calculation	1.5
Correct application of Bragg's law	1.5
Correct arithmetic	1
Correct final angles with units	1

Problem 11: Structure Factor of Diamond (6 points)

Solution: Diamond structure: FCC basis with second atom at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Structure factor:

$$F_{hkl} = f_{Si}[1 + e^{2\pi i(h/4+k/4+l/4)}] \times F_{FCC}$$

where $F_{FCC} = 1 + e^{\pi i(k+l)} + e^{\pi i(h+l)} + e^{\pi i(h+k)}$

F_{FCC} is zero when h, k, l are mixed.

When h, k, l are all even or all odd:

$$F_{hkl} = f_{Si}[1 + e^{2\pi i(h+k+l)/4}]F_{FCC}$$

For $h + k + l = 4n + 2$:

$$e^{2\pi i(4n+2)/4} = e^{\pi i(2n+1)} = -1$$

Therefore $1 + e^{2\pi i(h+k+l)/4} = 1 - 1 = 0$

Hence $F_{hkl} = 0$ when $h + k + l = 4n + 2$.

Grading Rubric:

Criterion	**Points**
Correct diamond structure basis	1
Correct expression for F_{hkl}	1.5
Correct F_{FCC} expression	1
Correct analysis of F_{FCC} zeros	0.5
Correct analysis of second atom contribution	1
Correct final condition ($h + k + l = 4n + 2$)	1

Problem 12: (100) Absence in BCC (4 points)

Solution: For BCC: basis atoms at (0,0,0) and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

Structure factor:

$$F_{100} = f[1 + e^{\pi i(1+0+0)}] = f[1 + e^{\pi i}] = f[1 - 1] = 0$$

The (100) reflection is absent because the waves scattered from the corner atoms and the body center atom are out of phase by π, causing destructive interference.

For SC with one atom basis at (0,0,0):

$$F_{100} = f \neq 0$$

Thus the (100) reflection is present in SC but absent in BCC.

Grading Rubric:

Criterion	**Points**
Correct BCC structure factor expression	1.5
Correct evaluation of $F_{100} = 0$	1
Correct SC structure factor evaluation	0.5
Physical explanation	1

Lecture 3 Problems

Problem 13: Powder Pattern Indexing for Iron (6 points)

Solution: For BCC, allowed reflections require $h + k + l$ even.
Using $\sin^2\theta = \frac{\lambda^2}{4a^2}(h^2 + k^2 + l^2)$
For BCC, the first allowed reflections:

(hkl)	$h^2 + k^2 + l^2$	$\sin^2\theta$
(110)	2	$\frac{\lambda^2}{4a^2} \times 2$
(200)	4	$\frac{\lambda^2}{4a^2} \times 4$
(211)	6	$\frac{\lambda^2}{4a^2} \times 6$
(220)	8	$\frac{\lambda^2}{4a^2} \times 8$
(310)	10	$\frac{\lambda^2}{4a^2} \times 10$
(222)	12	$\frac{\lambda^2}{4a^2} \times 12$

First five lines: (110), (200), (211), (220), (310)
Grading Rubric:

Criterion	**Points**
Correct BCC selection rule ($h + k + l$ even)	1
Correct calculation of $h^2 + k^2 + l^2$ for each reflection	2
Correct order of reflections	2
Correct identification of first five lines	1

Problem 14: Why Powder Method Works (4 points)

Solution: The powder method works because:

1. The powder contains crystallites in all possible orientations
2. For any family of planes (hkl), there will always be some crystallites oriented at the correct Bragg angle θ
3. Each set of planes produces a cone of diffracted radiation
4. The random orientation ensures that all allowed reflections appear in the pattern

Grading Rubric:

Criterion	**Points**
Random orientation of crystallites	1
Some crystallites always satisfy Bragg's law	1
Cone formation explanation	1
Complete coverage of all reflections	1

Problem 15: Multiplicity Factor for 110 (4 points)

Solution: For 110 planes in cubic crystals, the equivalent planes are obtained by permutations and sign changes of the indices.

Number of permutations of (1,1,0) with two different indices $= \frac{3!}{2!} = 3$

For each permutation, sign changes:

- For indices with no zeros: 2^3 sign combinations possible
- But (hkl) and $(\bar{h}\bar{k}\bar{l})$ are the same plane, so divide by 2

Multiplicity $= 3 \times 2^2 = 12$ (since the zero index has only 1 sign)

Alternative: 110 has 12 equivalent planes.

Grading Rubric:

Criterion	Points
Correct number of permutations	1.5
Correct treatment of sign changes	1.5
Correct final multiplicity (12)	1

Problem 16: 5-fold Rotation Incompatibility (5 points)

Solution: Consider a lattice row with spacing a. A C_n rotation by angle $2\pi/n$ maps lattice points to lattice points.

After rotation, the projection of the rotated vector onto the original row must be an integer multiple of a:

$$a\cos(2\pi/n) = ma \implies \cos(2\pi/n) = m$$

where m is an integer.

For $n = 5$: $\cos(72°) = 0.3090$, not an integer.

Therefore 5-fold rotation is incompatible with translational symmetry.

The only possible values are:

- $n = 1$: $\cos(360°) = 1$
- $n = 2$: $\cos(180°) = -1$
- $n = 3$: $\cos(120°) = -1/2$
- $n = 4$: $\cos(90°) = 0$
- $n = 6$: $\cos(60°) = 1/2$

Grading Rubric:

Criterion	Points
Correct setup with lattice spacing a	1
Correct condition $\cos(2\pi/n)$ must be integer	1.5
Correct evaluation for $n = 5$	1
Correct listing of allowed values	1.5

Problem 17: Point Group of Cube (3 points)

Solution: A cube has point group O_h (Schoenflies notation) or $m\bar{3}m$ (International notation).

Operations: 48 total

- 24 proper rotations (including identity)
- 24 improper rotations (including inversion)

Grading Rubric:

Criterion	Points
Correct Schoenflies notation (O_h)	1
Correct International notation ($m\bar{3}m$)	1
Correct number of operations (48)	1

Problem 18: Space Groups in 2D (2 points)

Solution: There are 17 space groups in 2D.

Grading Rubric:

Criterion	Points
Correct answer (17)	2

Lecture 4 Problems

Problem 19: Copper Properties (9 points)

Solution: (a) In FCC, atoms touch along face diagonal: $4r = \sqrt{2}a$

$$r = \frac{\sqrt{2}a}{4} = \frac{\sqrt{2} \times 3.61}{4} = 1.28 \text{ Å}$$

(b) Density:

$$\rho = \frac{4M}{N_A a^3}$$

$$= \frac{4 \times 63.55 \text{ g/mol}}{6.022 \times 10^{23} \text{ mol}^{-1} \times (3.61 \times 10^{-8} \text{ cm})^3}$$

$$= \frac{254.2}{6.022 \times 10^{23} \times 4.70 \times 10^{-23}} = \frac{254.2}{28.3} = 8.96 \text{ g/cm}^3$$

(c) $d_{111} = \frac{a}{\sqrt{3}} = \frac{3.61}{\sqrt{3}} = 2.08$ Å

Grading Rubric:

Criterion	Points
(a) Correct relationship and calculation of r	3
(b) Correct formula with correct values	2
(b) Correct arithmetic and units	1
(c) Correct formula and calculation	2
(c) Correct units	1

Problem 20: BCC Diffraction (6 points)

Solution: Given: $2\theta = 30°$, $\lambda = 1.54$ Å, (110) planes
$\theta = 15°$
Bragg's law: $2d_{110}\sin\theta = \lambda$

$$d_{110} = \frac{\lambda}{2\sin\theta} = \frac{1.54}{2\sin(15°)} = \frac{1.54}{2 \times 0.2588} = \frac{1.54}{0.5176} = 2.98 \text{ Å}$$

For cubic: $d_{110} = \frac{a}{\sqrt{1^2+1^2+0^2}} = \frac{a}{\sqrt{2}}$
Therefore: $a = d_{110}\sqrt{2} = 2.98 \times 1.414 = 4.21$ Å
Grading Rubric:

Criterion	Points
Correct Bragg's law application	2
Correct calculation of d_{110}	2
Correct relationship between a and d_{110}	1
Correct final value with units	1

Problem 21: 2D Hexagonal Lattice (6 points)

Solution: Given: $\mathbf{a}_1 = a(1, 0)$, $\mathbf{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$
The reciprocal lattice vectors have already been derived in Problem 7:

$$\mathbf{b}_1 = \frac{2\pi}{a}(1, -\frac{1}{\sqrt{3}}), \quad \mathbf{b}_2 = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}})$$

Lengths:

$$|\mathbf{b}_1| = \frac{2\pi}{a}\sqrt{1 + \frac{1}{3}} = \frac{2\pi}{a}\frac{2}{\sqrt{3}}$$

$$|\mathbf{b}_2| = \frac{2\pi}{a}\frac{2}{\sqrt{3}}$$

The angle between $\mathbf{b}_1$ and $\mathbf{b}_2$:

$$\cos\theta = \frac{\mathbf{b}_1 \cdot \mathbf{b}_2}{|\mathbf{b}_1||\mathbf{b}_2|} = \frac{(\frac{2\pi}{a})(-\frac{1}{\sqrt{3}}) \times (\frac{2\pi}{a})(\frac{2}{\sqrt{3}})}{(\frac{2\pi}{a}\frac{2}{\sqrt{3}})^2} = \frac{1}{2}$$

Therefore $\theta = 60°$, confirming hexagonal symmetry.
Grading Rubric:

Criterion	Points
Correct $\mathbf{b}_1$ and $\mathbf{b}_2$	2
Correct lengths calculation	1
Correct angle calculation	2
Conclusion: reciprocal lattice is hexagonal	1

Problem 22: Sodium BCC (6 points)

Solution: Given: $2\theta = 29.8°$, $\lambda = 1.54$ Å

$\theta = 14.9°$

Bragg's law: $2d\sin\theta = \lambda$

$$d = \frac{\lambda}{2\sin\theta} = \frac{1.54}{2\sin(14.9°)} = \frac{1.54}{2 \times 0.2571} = \frac{1.54}{0.5142} = 2.99 \text{ Å}$$

For BCC, first reflection is from (110) planes.

$$d_{110} = \frac{a}{\sqrt{2}}$$

Therefore: $a = d_{110}\sqrt{2} = 2.99 \times 1.414 = 4.23$ Å

The reflecting planes are (110).

Grading Rubric:

Criterion	Points
Correct Bragg's law application	2
Correct calculation of d	1.5
Correct identification of BCC first reflection	1
Correct a calculation	1
Correct identification of planes	0.5

Problem 23: Zinc Blende Structure Factor (7 points)

Solution: Zinc blende: FCC lattice with basis of two different atoms at (0,0,0) and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$.

Structure factor:

$$F_{hkl} = f_{Zn}F_{FCC} + f_S e^{2\pi i(h+k+l)/4}F_{FCC}$$

where $F_{FCC} = 1 + e^{\pi i(k+l)} + e^{\pi i(h+l)} + e^{\pi i(h+k)}$

$$F_{hkl} = F_{FCC}[f_{Zn} + f_S e^{2\pi i(h+k+l)/4}]$$

Systematic absences: 1. When h, k, l are mixed (FCC structure factor zero): $F_{hkl} = 0$
2. When h, k, l are all even or all odd:

$$F_{hkl} = F_{FCC}[f_{Zn} + f_S e^{2\pi i(h+k+l)/4}]$$

Additional absences occur when $f_{Zn} + f_S e^{2\pi i(h+k+l)/4} = 0$

Grading Rubric:

Criterion	Points
Correct zinc blende basis	1
Correct F_{FCC} expression	1.5
Correct combined structure factor expression	2
Correct mixed indices condition	1
Correct additional absence condition	1.5

Problem 24: Neumann's Principle Application (5 points)

Solution: Neumann's principle states that the symmetry elements of any physical property must include the symmetry elements of the point group of the crystal.

For a cubic crystal with inversion symmetry (e.g., NaCl):

- Inversion symmetry requires any physical property to be invariant under $\mathbf{r} \to -\mathbf{r}$
- Piezoelectricity requires a change in polarization under strain: $P_i = d_{ijk}\sigma_{jk}$
- Under inversion, $P \to -P$ and $\sigma \to \sigma$ (stress is a polar tensor)
- Therefore $d_{ijk} = -d_{ijk} \implies d_{ijk} = 0$

For zinc blende (cubic but no inversion center):

- The symmetry operations do not include inversion
- Therefore d_{ijk} can be non-zero
- Zinc blende is indeed piezoelectric

Grading Rubric:

Criterion	Points
Correct statement of Neumann's principle	1
Correct analysis of inversion symmetry	1.5
Correct analysis of piezoelectric tensor	1.5
Correct conclusion for zinc blende	1

3 Total Points Summary

4 TA Grading Guidelines

4.1 General Principles

1. **Consistency:** Apply the same standards to all student submissions.
2. **Partial Credit:** Award partial credit for correct intermediate steps.
3. **Units:** Deduct 0.5 points for missing or incorrect units.
4. **Significant Figures:** Deduct 0.5 points for significant figure errors.
5. **Clarity:** Deduct up to 1 point for unclear or disorganized solutions.

Problem	Topic	Points
1	FCC Packing Fraction	5
2	Diamond Atoms	3
3	(110) Plane in BCC	4
4	Miller Indices	4
5	Interplanar Spacing	4
6	$(11\bar{2})$ Plane	4
7	2D Hexagonal Reciprocal	8
8	BCC $\leftrightarrow$ FCC	8
9	$\mathbf{G} \perp$ Plane	6
10	Bragg Angle	5
11	Diamond Structure Factor	6
12	(100) Absence	4
13	Powder Pattern Indexing	6
14	Powder Method Explanation	4
15	Multiplicity Factor	4
16	5-fold Rotation	5
17	Cube Point Group	3
18	2D Space Groups	2
19	Copper Properties	9
20	BCC Diffraction	6
21	2D Hexagonal (Repeated)	6
22	Sodium BCC	6
23	Zinc Blende Structure	7
24	Neumann's Principle	5
Total		**124**

4.2 Common Mistakes by Topic

4.2.1 Crystal Structures

- Mixing up FCC and BCC coordination numbers
- Incorrect number of atoms per unit cell
- Wrong packing fraction formulas

4.2.2 Miller Indices

- Forgetting to take reciprocals
- Not clearing fractions properly
- Confusing families ($\{hkl\}$) with specific planes (hkl)

4.2.3 Reciprocal Lattice

- Incorrect application of $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$
- Wrong cross product calculations
- Forgetting the 2π factor

4.2.4 Diffraction

- Confusing Bragg angle θ with 2θ
- Incorrect structure factor calculations
- Missing systematic absence conditions

4.3 Feedback Template

For efficient grading, use the following template for feedback:

```
Problem X:
- Approach: [Excellent/Good/Needs Improvement]
- Calculation: [Correct/Minor errors/Major errors]
- Units: [Correct/Incorrect]
- Score: [X/Y points]

Comments: [Specific feedback]
```

5 Supplementary Teaching Materials

5.1 Key Formulas Quick Reference

$$\begin{aligned}
\text{Lattice: } & \mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3 \\
\text{Reciprocal: } & \mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3 \\
\text{Orthogonality: } & \mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij} \\
\text{Bragg's Law: } & 2d\sin\theta = n\lambda \\
\text{Interplanar: } & d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}} \\
\text{Structure Factor: } & F_{hkl} = \sum_j f_j e^{2\pi i(hx_j+ky_j+lz_j)}
\end{aligned}$$

5.2 Structure Factor Rules Table

Structure	Reflection Conditions
SC	All (hkl)
BCC	$h+k+l$ even
FCC	h, k, l all even or all odd
Diamond	h, k, l all even or all odd, and $h+k+l$ not $4n+2$
NaCl	h, k, l all even or all odd
CsCl	All (hkl)